2.10 Organic chemistry

Related topics in *Units 4* and *5* will assume knowledge of this material.

During this topic there will be the opportunity to carry out a number of internal assessment activities. Please see *Appendix 9* for more details.

Students will be assessed on their ability to:

1 Alcohols	a give examples of, and recognise, molecules that contain the alcohol functional group.
	b demonstrate an understanding of the nomenclature and corresponding structural, displayed and skeletal formulae of alcohols, and classify them as primary, secondary or tertiary

c describe the following chemistry of alcohols:

   i combustion

   ii reaction with sodium

   iii substitution reactions to form halogenoalkanes, including reaction with PCl_5 and its use as a qualitative test for the presence of the –OH group

   iv oxidation using potassium dichromate(VI) in dilute sulfuric acid on primary alcohols to produce aldehydes and carboxylic acids and on secondary alcohols to produce ketones

d demonstrate an understanding of, and practise, the preparation of an organic liquid (reflux and distillation), eg oxidation of alcohols.

2 Halogeno alkanes

a demonstrate an understanding of the nomenclature and corresponding structural, displayed and skeletal formulae for halogenoalkanes, including the distinction between primary, secondary and tertiary structures

b interpret given data and observations comparing the reactions and reactivity of primary, secondary and tertiary compounds

c carry out the preparation of an halogenoalkane from an alcohol and explain why a metal halide and concentrated sulfuric acid should not be used when making a bromoalkane or an iodoalkane

d describe the typical behaviour of halogenoalkanes. This will be limited to treatment with:

   i aqueous alkali, eg KOH (aq)

   ii alcoholic potassium hydroxide

   iii water containing dissolved silver nitrate

   iv alcoholic ammonia

e carry out the reactions described in 2.10.2d i, ii, iii

f discuss the uses of halogenoalkanes, eg as fire retardants and modern refrigerants.